

MCS-1 Tutorial (Quiz-1 Solutions)

Problem 1. Let $X_1, \dots, X_n$ be independent random variables with common mean μ and variance σ^2 . Let

$$Y = \frac{1}{n} \sum_{i=1}^n X_i.$$

Show that

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - Y)^2 \right] = (n-1)\sigma^2.$$

Problem 2. Let $X \sim \text{Geometric}(p)$ and $M > 0$ be integer. Define $Z = \min(X, M)$. Compute $\mathbb{E}[Z]$.

Problem 3. A stock market trader buys 100 shares of stock A and 200 shares of stock B . Let X, Y be the price changes of A and B over a certain period, and assume that the joint PMF of X and Y is uniform over the set of integers x and y satisfying

$$-2 \leq x \leq 4, \quad -1 \leq y - x \leq 1$$

- Find marginals and means of X, Y .
- Find expected profit.

Problem 4. Let X and Y be independent random variables having geometric densities with parameters p_1 and p_2 respectively. Find

- $P(X \geq Y)$
- $P(X = Y)$

Problem 5. Alvin's database of friends contains n entries, but due to a software glitch, the addresses correspond to the names in a totally random fashion. Alvin writes a holiday card to each of his friends and sends it to the (software-corrupted) address. What is the probability that at least one of his friends will get the correct card? Hint: Use the inclusion-exclusion formula.

